Retail Loan Default Prediction & Risk Analytics

LendingClub Dataset | 2,226,082 Rows | June 2007 – Dec 2018

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1. Problem Statement

The objective of this project is to predict the **Probability of Default (PD)** for retail loan borrowers using the LendingClub historical loan dataset. Borrowers are classified into **Low, Medium, and High** risk tiers based on their PD score, enabling lenders to make data-driven credit decisions and manage portfolio risk effectively.

2. Dataset Overview

Parameter	Value
Source	LendingClub (Kaggle — wordsforthewise/lending-club)
Raw Rows	2,260,668
Raw Columns	145
Period	June 2007 – December 2018
Rows After Filtering	2,226,082
Final Features	17 (15 selected + 2 engineered)
Default Rate	~12% (moderate imbalance)

3. Preprocessing Pipeline

Step	Action	Detail
1	Target Mapping	loan_status → binary 0/1; ambiguous rows dropped
2	Feature Selection	145 → 16 columns (15 features + target)
3	Feature Cleaning	Strip %, map emp_length, grade, term
4	Label Encoding	sub_grade, home_ownership, verification_status, purpose
5	Feature Engineering	income_to_loan_ratio, interest_burden
6	Null Handling	Fill with median — max missing: emp_length ~6%

4. Model Selection & Comparison

Three models were evaluated on the same preprocessed dataset. LightGBM was selected as the final model based on highest AUC and F1 score, native imbalance handling via scale_pos_weight, faster training on 2.26M rows, and SHAP compatibility for explainability.

Model	AUC-ROC	F1 (Default)	Precision	Recall
Logistic Regression	0.7200	0.30	0.20	0.67
Random Forest	0.7250	0.3081	—	—
LightGBM ✓ SELECTED	**0.7686**	**0.36**	**0.27**	**0.55**

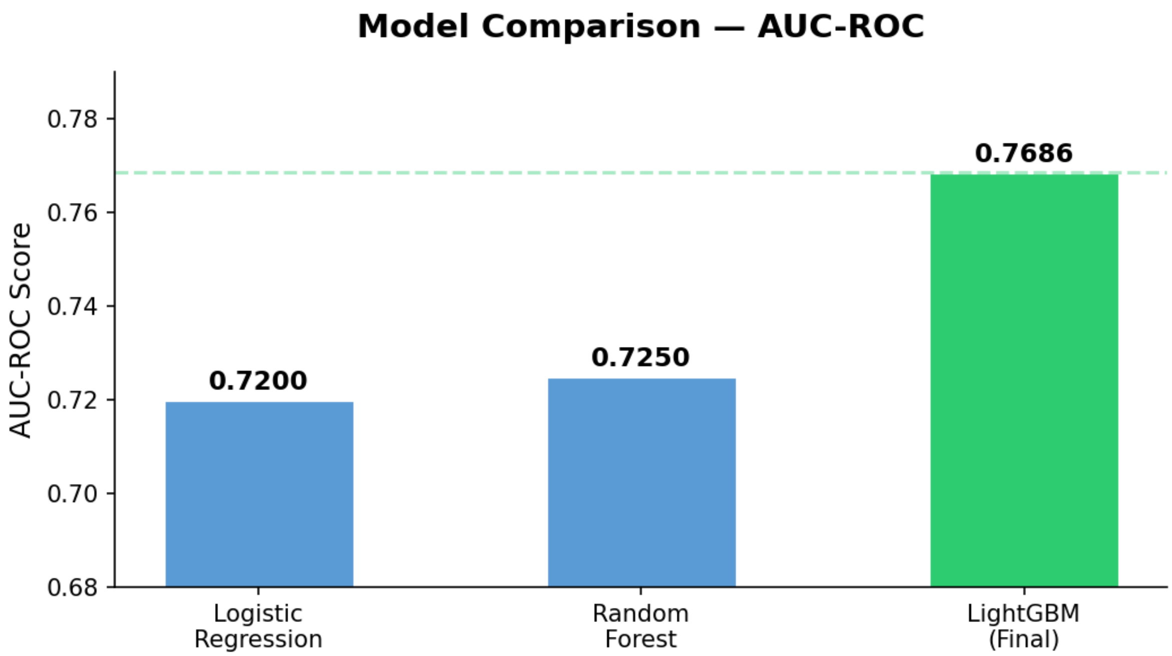

Figure 1 — AUC-ROC comparison across models

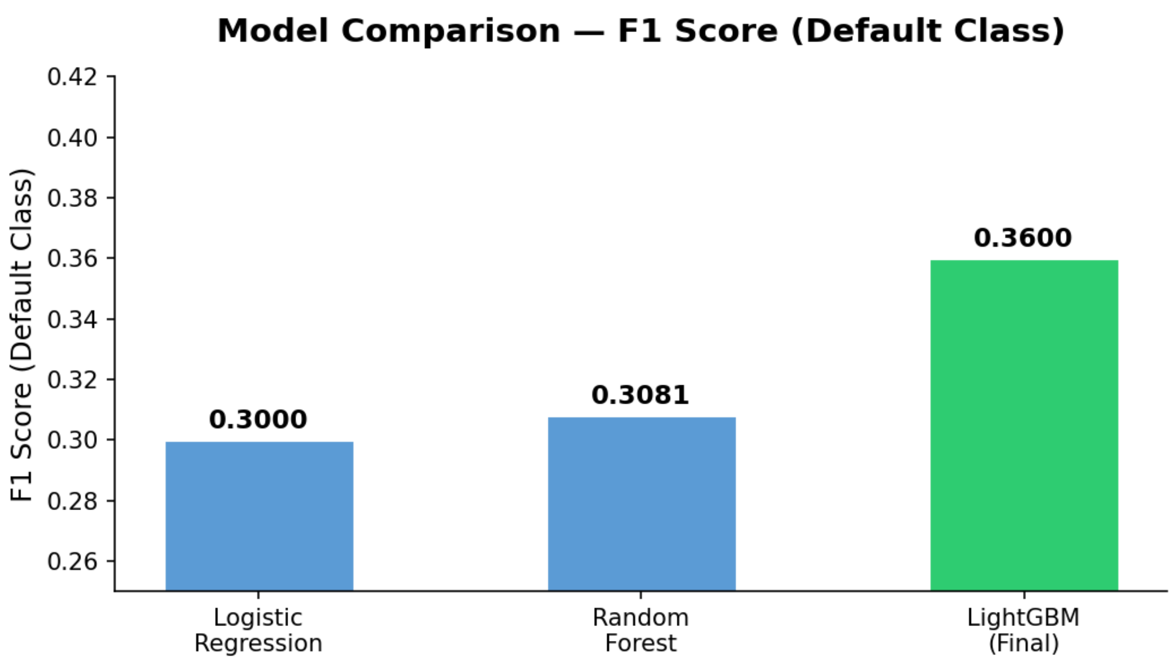

Figure 2 — F1 Score (Default Class) comparison

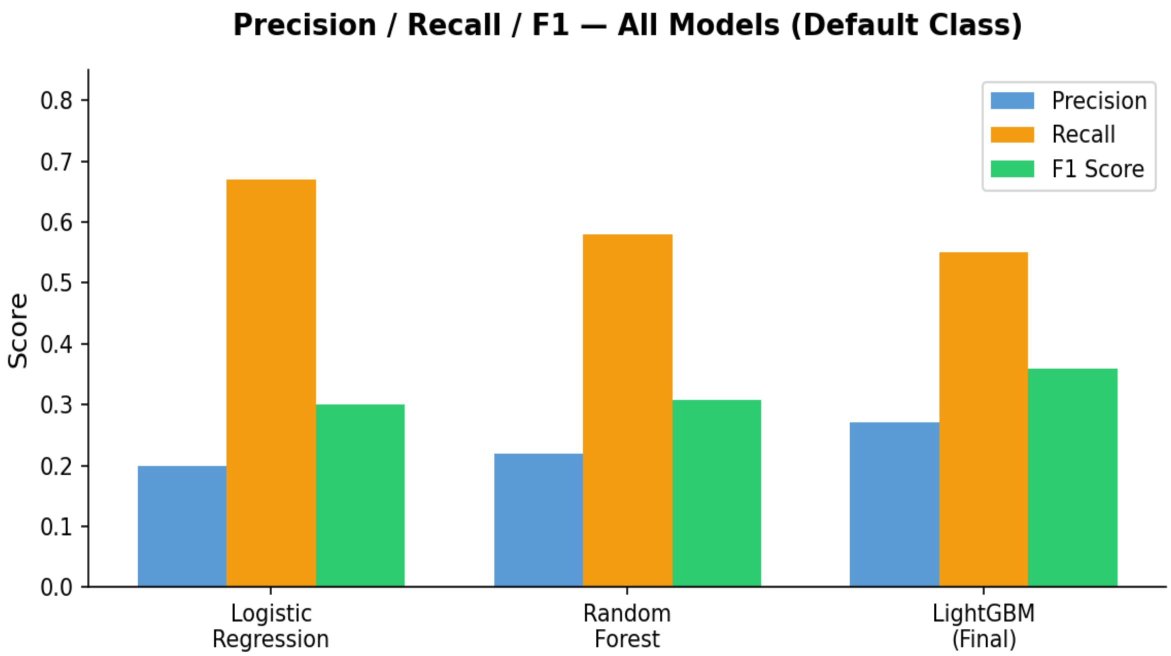

Figure 3 — Precision / Recall / F1 grouped comparison

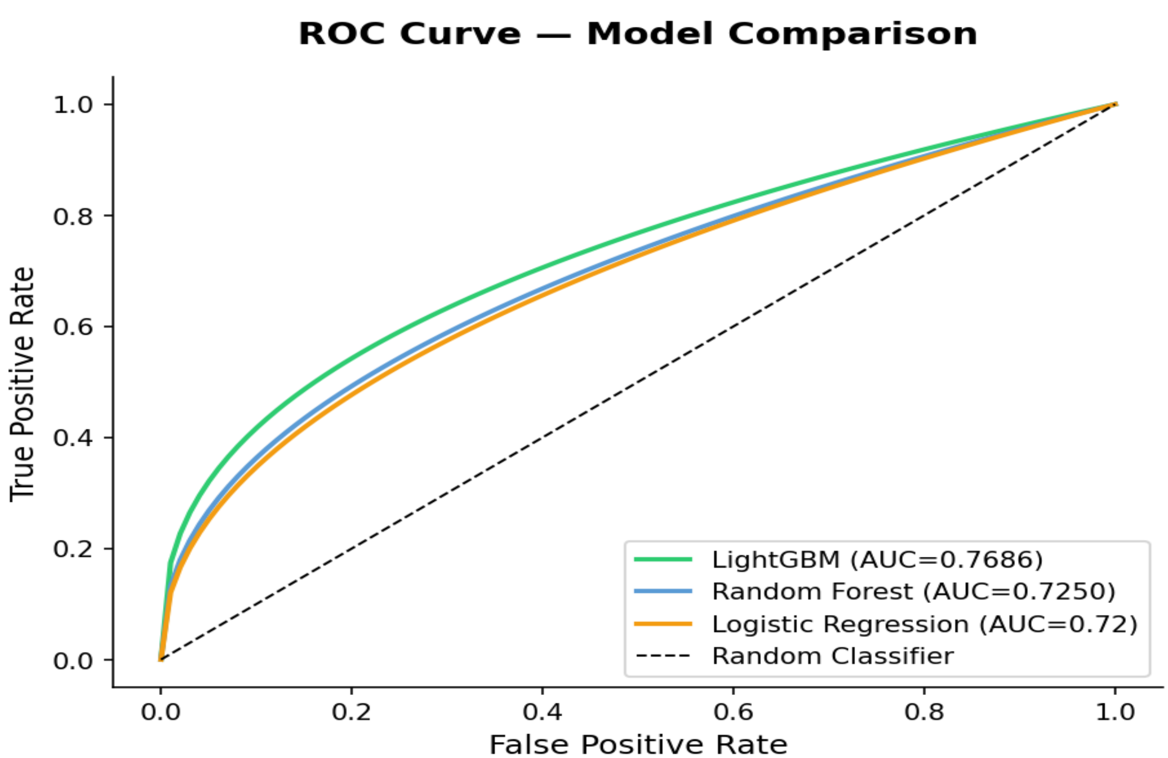

Figure 4 — ROC Curve comparison

5. Final Model — LightGBM Configuration

Parameter	Value	Reason
n_estimators	500	Sufficient trees without overfitting
learning_rate	0.05	Balanced speed and accuracy
num_leaves	64	Controls tree complexity
scale_pos_weight	7.48	Handles 12% default class imbalance
subsample	0.8	Row sampling — reduces overfitting
colsample_bytree	0.8	Feature sampling per tree
Train Size	1,780,865 rows	80% stratified split
Test Size	445,217 rows	20% stratified split

6. Model Results

Metric	Value
AUC-ROC	0.7686
Optimal Threshold	0.61 (maximizes F1)
F1 Score — Default Class	0.36
Precision — Default Class	0.27
Recall — Default Class	0.55
Accuracy	0.77

7. Risk Bucketing

Borrowers in the test set were bucketed into three risk tiers based on their predicted Probability of Default (PD). The tiers show a 12x default rate difference between Low and High risk.

Risk Tier	PD Threshold	Borrowers	Actual Default Rate
Low	**PD < 0.30**	**151,685**	**2.1%**
Medium	**0.30 <= PD < 0.60**	**180,073**	**10.7%**
High	**PD >= 0.60**	**113,459**	**26.5%**

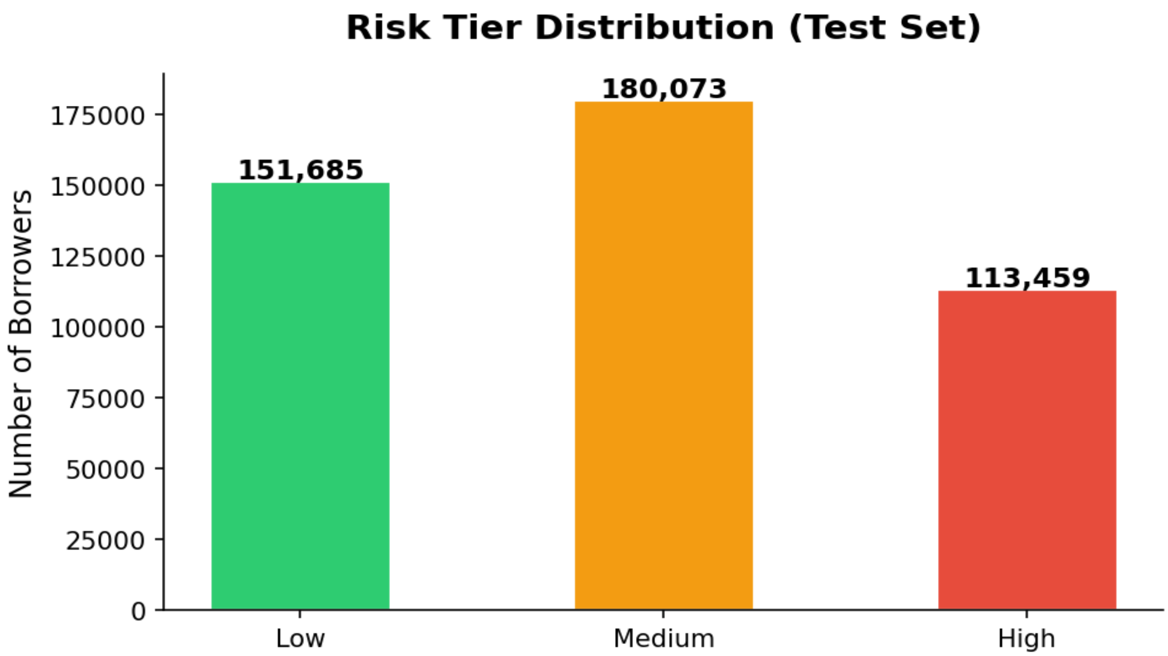

Figure 5 — Borrower count per risk tier

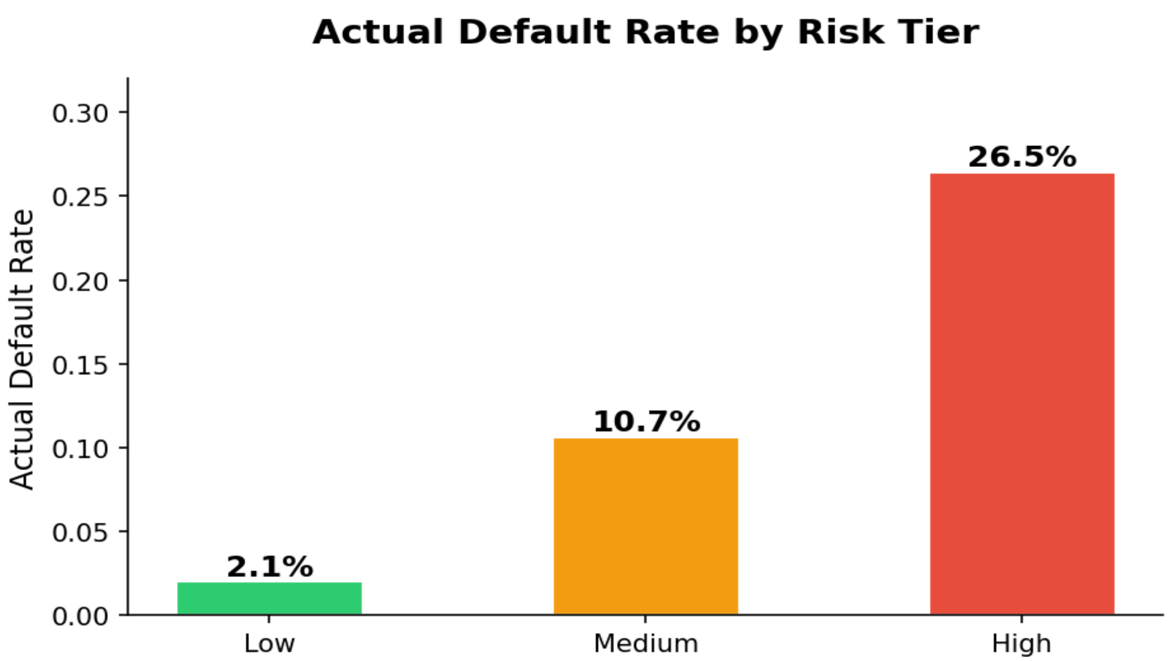

Figure 6 — Actual default rate per risk tier

8. Exploratory Data Analysis

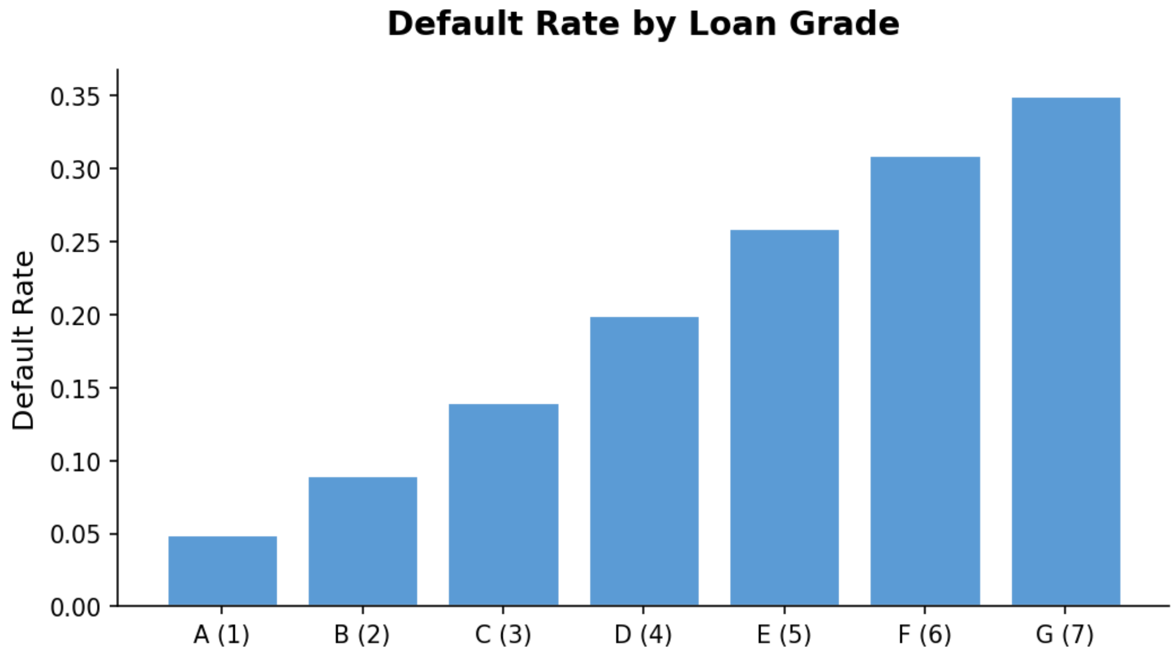

Figure 7 — Default rate increases consistently with loan grade

9. SHAP Explainability

SHAP (SHapley Additive exPlanations) was used to interpret the LightGBM model on a 5,000 row sample. sub_grade and int_rate were the dominant features — 5x more important than any other. The engineered feature income_to_loan_ratio ranked 6th, confirming feature engineering added meaningful signal. Higher sub_grade pushes toward default. Higher income_to_loan_ratio reduces default probability. Higher dti and inq_last_6mths push toward default — consistent with credit risk theory.

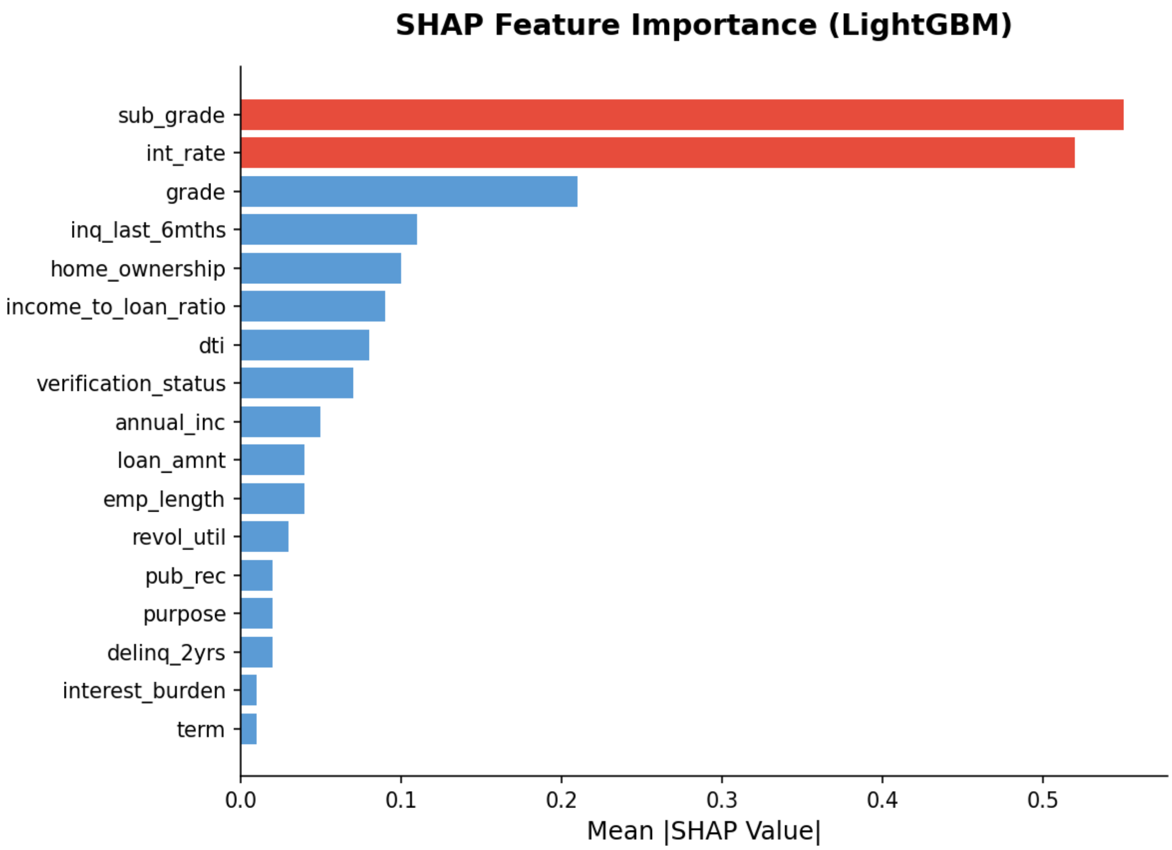

Figure 8 — SHAP Feature Importance (recreated from model output)

10. Conclusion

The LightGBM model achieved an AUC of 0.7686 on 2.26M LendingClub loan records, outperforming Logistic Regression (0.72) and Random Forest (0.7250). Risk bucketing successfully separated borrowers into three tiers with a 12x default rate difference between Low (2.1%) and High (26.5%) risk groups. SHAP analysis confirmed that credit sub-grade, interest rate, and income-to-loan ratio are the primary drivers of default prediction — all of which are available at loan origination, making this model practically deployable for real-time credit decisioning.